

SUMMARY

An aspiring graduate student with experience in web development, data processing and machine learning, skilled in building software solutions and optimizing data workflows. A proficient team player with scrum master experience and strong communication skills. Actively looking for full-time Software Engineering position.

- **Programming Languages:** Python, TypeScript, Java, C/C++, HTML, CSS, Scala
- **Software:** Git, NestJS, React, MongoDB, SQL, Docker, Kubernetes, Terraform, Apache Spark, Jupyter Notebook

EDUCATION

Carnegie Mellon University Expected May 2025
Master of Science in Computational Data Science GPA: 4.00/4.00

Relevant Courses: Cloud Computing, Machine Learning, Interactive Data Science, LLM: Methods and Applications

New York University Sept 2019-May 2023
Bachelor of Arts in Computer and Data Science GPA: 3.93/4.00

Relevant Courses: Basic Algorithms, Data Structures, Object Oriented Programming, Operating Systems, Natural Language Processing, Data Management and Analysis, Causal Inference, Deep Learning, Processing Big Data for Analytics Applications

WORK EXPERIENCE

The Trade Desk | *Data Scientist Intern* | Denver, CO May 2024-Aug 2024

- Implemented a scalable XGBoost model to optimize Connected TV ad targeting, enhancing model efficiency and system integration
- Engineered automated data pipelines with Airflow, AWS EMR, and Kubernetes, ensuring proactive monitoring for data/model drift
- Conducted large-scale data processing and feature engineering in Databricks, focusing on performance and resource optimization
- Achieved a 40% improvement in model precision and AUROC, significantly enhancing real-time decision-making

Legion AI | *Software Engineer Intern* | Pittsburgh, PA Dec 2023-Feb 2024

- Implemented REST API with NestJS to maintain live conversation sessions with Large Language Model (LLM) Endpoint, utilized prompt compression with Context Summary and Vector Database embedding to bypass context window limitation
- Adopted DevOps methodology by automating the build and deployment of Docker images of applications and Helm charts to the Kubernetes clusters using GitHub Actions to realize continuous integration in the development process
- Orchestrated elasticity policies to maintain the Quality of Service and reduce cost of web services on AWS Auto Scaling Group by analyzing workload patterns on CloudWatch, using Terraform as part of the deployment process for infrastructure automation

PROJECTS

Conversational Business Intelligence with LLMs for Mining | Komatsu & Carnegie Mellon University Sep 2024-Dec 2024

- Collaborated with Komatsu to design and implement a Conversational Business Intelligence system using LLMs, enabling non-technical stakeholders to seamlessly access mining machine status data
- Designed and deployed a web application with Streamlit, integrating an LLM Endpoint to translate natural language queries into SQL, retrieve relevant data, and provide human-readable responses
- Performed parameter-efficient fine-tuning of LLMs with diverse Text-to-SQL datasets, leveraging QLoRA and 4-bit quantization techniques to reduce costs
- Achieved over 88% accuracy in internal testing on daily data engineering queries, created developer-focused tools to streamline prompt engineering and facilitate efficient error analysis during model iteration

Optimizing Twitter Analytics Microservice | Carnegie Mellon University Mar 2024-May 2024

- Led the optimization of a cloud-based microservices architecture for a Twitter analytics web service, improving throughput and cost-efficiency across three phases by leveraging managed services and advanced database schemas
- Spearheaded the transition to AWS fully-managed services, enhancing the performance/cost ratio by over 40% while ensuring scalability, fault tolerance, and seamless integration with existing web and storage tiers.
- Optimized database schemas and ETL processes, resulting in a 250% increase in requests per second (RPS) under heavy loads
- Implemented CI/CD pipelines and automated deployment with Terraform and Kubernetes, reducing deployment time and errors

Real-Time Data Processing for Ride-Hailing | Carnegie Mellon University Feb 2024-Apr 2024

- Designed and implemented a real-time data processing system using Apache Kafka and Apache Samza to manage and analyze large-scale GPS and IoT data streams for a ride-hailing service, enabling efficient driver-client matching and targeted advertising
- Developed Kafka producers to generate and partition data streams based on geographic location, and consumed these streams using Samza to calculate driver-client match scores with low-latency processing
- Deployed and managed the stream processing system on AWS, leveraging YARN clusters to ensure fault tolerance, scalability, and efficient resource utilization